

PROJECT REPORT
ON
DESIGN AND ANALYSIS OF 6T SRAM
CELL
USING CADENCE VIRTUOSO

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ABSTRACT

This project presents the design, simulation, and analysis of a 6-Transistor (6T) Static Random Access Memory (SRAM) cell using Cadence Virtuoso EDA tool. The 6T SRAM cell is implemented and characterized in both 90nm (gpdk090) and 180nm (gpdk180) CMOS technology nodes. Key aspects studied include Voltage Transfer Characteristics (VTC), Static Noise Margin (SNM) via butterfly curves, read/write transient analysis, power consumption with and without stacking effect, layout design, and integration with a tristate inverter-based Read-Write block and latch-type sense amplifier.

The stacking effect analysis demonstrates the reduction in subthreshold leakage current when multiple NMOS transistors are placed in series in the OFF state. A complete six-cycle read-write simulation is presented using the integrated Read-Write block, verifying correct initialisation, write-1, hold, pre-charge, and read operations. The transient waveforms for PRE, EN, D, WL, BL, BLB, Q, and QB are presented and analysed. Comparative power results between 90nm and 180nm nodes are tabulated and discussed.

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1. INTRODUCTION

Static Random Access Memory (SRAM) is a type of semiconductor memory that uses bistable latching circuitry to store each bit. Unlike Dynamic RAM (DRAM), SRAM does not need to be periodically refreshed, making it faster and more suitable for cache memory, register files, and embedded memory applications in modern digital systems.

The 6-Transistor (6T) SRAM cell is the industry-standard topology, consisting of two cross-coupled CMOS inverters forming a bistable latch (storage nodes Q and QB), and two NMOS access transistors controlled by a Wordline (WL) connecting the cell to the Bitlines (BL and BLB). This project explores the design and characterisation of the 6T SRAM cell in both 90nm and 180nm CMOS technology nodes using Cadence Virtuoso.

Key contributions of this project are:

- Schematic design in Cadence Virtuoso using gpdk090 (90nm) and gpdk180 (180nm)
- VTC and butterfly curve analysis for Static Noise Margin (SNM) verification
- Stacking vs. non-stacking leakage power analysis in 90nm
- Read operation with precharge circuit in 90nm
- Write and transient analysis in 180nm
- Layout design and verification in 180nm
- Integration with tristate inverter-based Read-Write block and full read-write simulation
- Latch-type sense amplifier design in 90nm
- Comparative power analysis between 90nm and 180nm

2. BACKGROUND AND THEORY

2.1 6T SRAM Cell — Structure and Operation

The 6T SRAM cell comprises two PMOS load transistors (PM0, PM2), two NMOS driver transistors (NM0, NM4/NM5), and two NMOS access transistors connected to BL and BLB. The cross-coupled inverter pair forms the storage latch. During HOLD (WL=0), the access transistors are off and the cell retains its state. During WRITE (WL=1), the bitline drivers overpower the cell feedback to force a new state. During READ (WL=1, bitlines precharged), the stored charge creates a differential on BL and BLB that is sensed by an amplifier.

2.2 VTC and Static Noise Margin (SNM)

The Voltage Transfer Characteristic (VTC) plots output voltage vs. input voltage for each inverter. Superimposing two mirrored VTCs produces the butterfly curve. The side length of the largest square inscribed in each lobe defines the Static Noise Margin (SNM) — a measure of hold-state stability. A larger SNM indicates a more robust cell.

2.3 Stacking Effect on Leakage Current

When two or more NMOS transistors are stacked in series and all are OFF, the intermediate node rises to a small positive voltage. This causes: (1) a negative VGS on the lower transistor, exponentially reducing subthreshold leakage; (2) negative VBS increasing effective threshold voltage via body effect; (3) reduced VDS across the upper transistor, lowering DIBL. Together, these reduce total leakage by 3x–10x compared to a single OFF transistor.

2.4 Tristate Inverter and Read-Write Block

A tristate inverter extends the standard CMOS inverter with additional enable transistors, allowing a high-impedance (Z) output when disabled. The Read-Write block combines a NOT-CMOS gate with a tristate inverter to generate BL and BLB control signals based on data input D, enable EN, and precharge PRE signals.

2.5 Latch-Type Sense Amplifier

The sense amplifier amplifies the small voltage differential between BL and BLB developed during a read operation. The latch-type sense amplifier uses cross-coupled inverters in positive feedback to rapidly regenerate the differential to full logic swing, activated by a sense enable signal.

3. DESIGN METHODOLOGY

3.1 Tool Used: Cadence Virtuoso

All schematics, symbols, simulations, and layout were designed using Cadence Virtuoso EDA platform with the Spectre simulator. Two PDKs were used:

- gpd090 — 90nm Generic Process Design Kit (VDD = 1.2V)
- gpd180 — 180nm Generic Process Design Kit (VDD = 1.8V)

3.2 Simulation Setup

Transient and DC analyses were used. For power, the Spectre calculator expression `average(getData("pwr"?result "tran"))` gives correct average transient power. Transistor sizing follows standard 6T SRAM design rules: Cell Ratio (CR = W/L of driver / W/L of access) > 1 for read stability, and Pull-up Ratio (PR = W/L of load / W/L of access) < 1 for write-ability.

4. 6T SRAM CELL — 90nm (gpdk090)

4.1 Schematic — 90nm

The 6T SRAM cell schematic was designed in Cadence Virtuoso using the gpdk090 library with VDD = 1.2V. The cell consists of two cross-coupled CMOS inverters forming the bistable storage latch (nodes Q and QB) and two NMOS access transistors connected to BL and BLB. PMOS transistors are sized $W=120\text{n}/L=100\text{n}$ and NMOS transistors $W=120\text{n}/L=100\text{n}$ as per the gpdk090 PDK specifications.

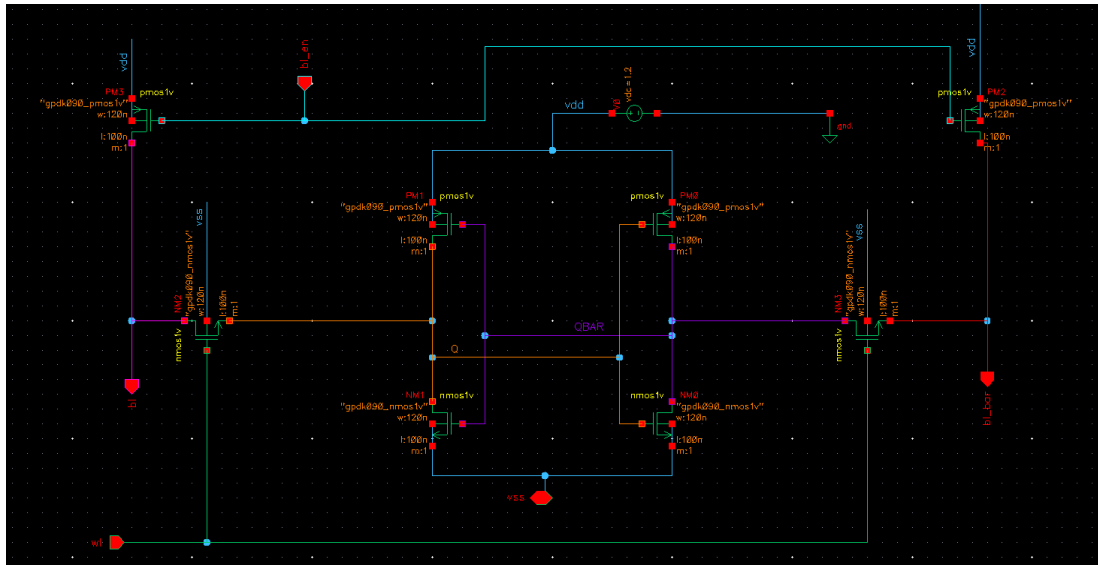


Figure 1: 6T SRAM Schematic with Precharge Circuit — 90nm (gpdk090)

The schematic includes a precharge PMOS transistor (PM3) connected to BL and controlled by bl_en, enabling BL to be precharged to VDD before each read operation. The storage nodes Q and QBAR are clearly labelled and the cross-coupling between the two inverter stages is visible.

4.2 Symbol — 6T SRAM

A hierarchical symbol for the 6T SRAM cell was created in Cadence Virtuoso Symbol Editor. The symbol exposes ports for BL, BLB, WL, VDD, GND, Q, and QB, enabling its use in larger hierarchical circuits such as the Read-Write block integration and sense amplifier testbench.

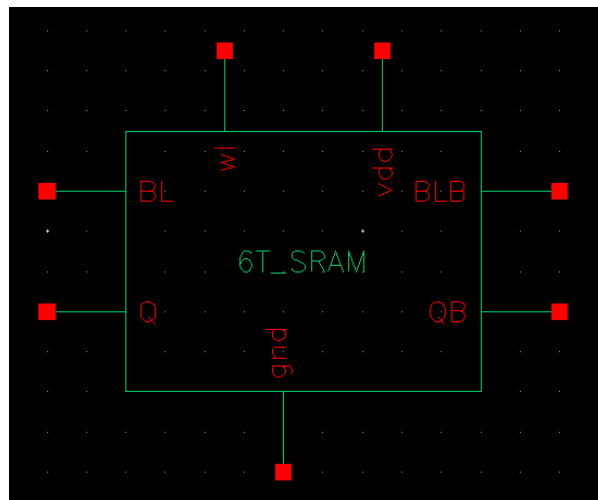


Figure 2: 6T SRAM Cell Symbol — Cadence Virtuoso

The symbol provides a clean interface for the 6T SRAM cell. All six ports (BL, BLB, WL on the signal side; VDD, GND for power; Q and QB as internal storage outputs) are accessible, making hierarchical instantiation straightforward.

4.3 Voltage Transfer Characteristic (VTC) — 90nm

The VTC was obtained by sweeping the input voltage of each inverter from 0V to 1.8V using a DC analysis. The resulting curve shows the high-gain inverting characteristic of the CMOS inverter, with a sharp transition around the switching threshold ($V_M \approx 0.7V$). The steep transition region confirms strong drive capability and good noise immunity.

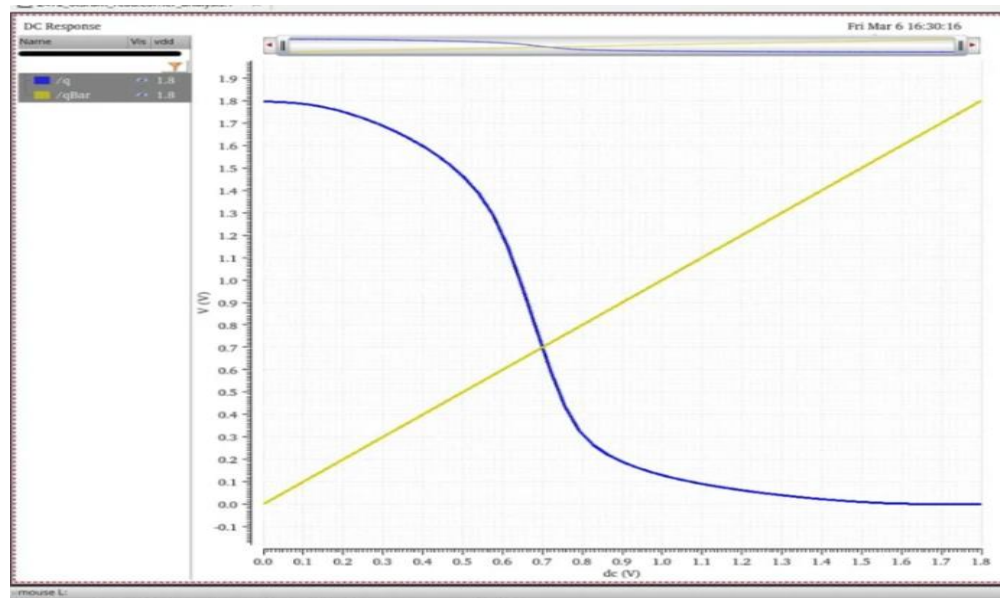


Figure 3: Voltage Transfer Characteristic (VTC) — 90nm gpd090

The VTC demonstrates a sharp high-gain transition near $V_M \approx 0.7V$. The output swings cleanly from VDD (1.8V) to 0V confirming full rail-to-rail operation. The steepness of the transition indicates a high voltage gain, which is critical for reliable bistable latch operation in the SRAM cell.

4.4 Butterfly Curve — 90nm

The butterfly curve is generated by superimposing the VTCs of both inverters (with one mirrored), producing the characteristic eye-shaped plot. The largest square that can be inscribed in each eye defines the Static Noise Margin (SNM). A positive SNM confirms stable bistable operation.

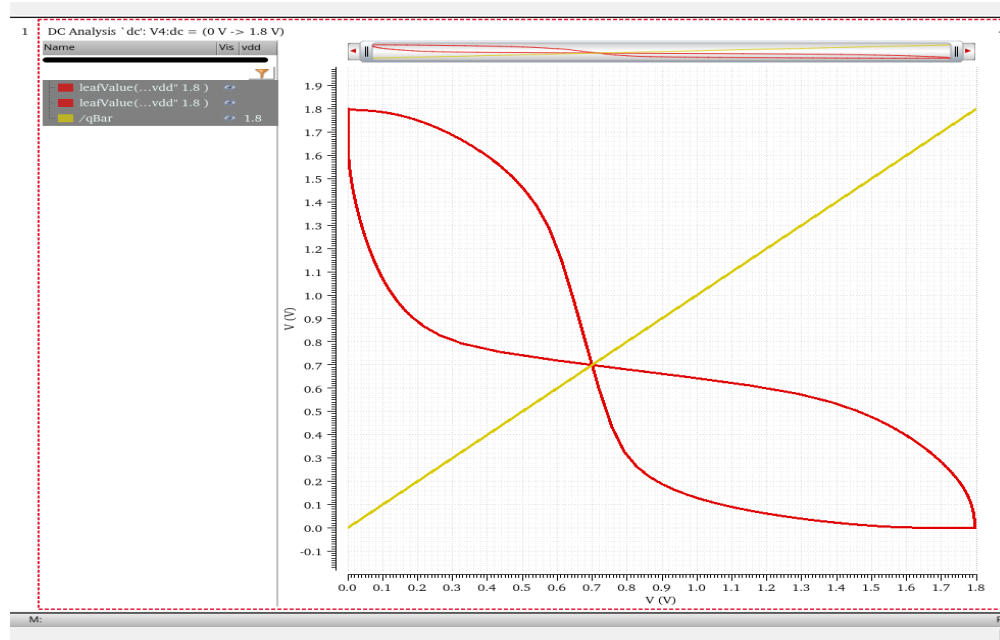


Figure 4: Butterfly Curve for SNM Analysis — 90nm gpd090

The butterfly curve shows two clearly defined stable operating points ($Q=0, QB=1$ and $Q=1, QB=0$) and one unstable equilibrium. The SNM squares inscribed in each eye confirm a positive noise margin, verifying that the cell can reliably hold its stored value in the absence of a write operation. The intersection of the two VTCs at the diagonal confirms correct sizing.

4.5 Read Operation with Precharge — 90nm

The precharge circuit charges BL and BLB to VDD before each read cycle. When WL is asserted, the stored value causes one bitline to discharge through the access and driver transistors while the other remains near VDD. The transient waveform captures bl_bar, wl, bl, bl_en, QBAR, and Q signals.

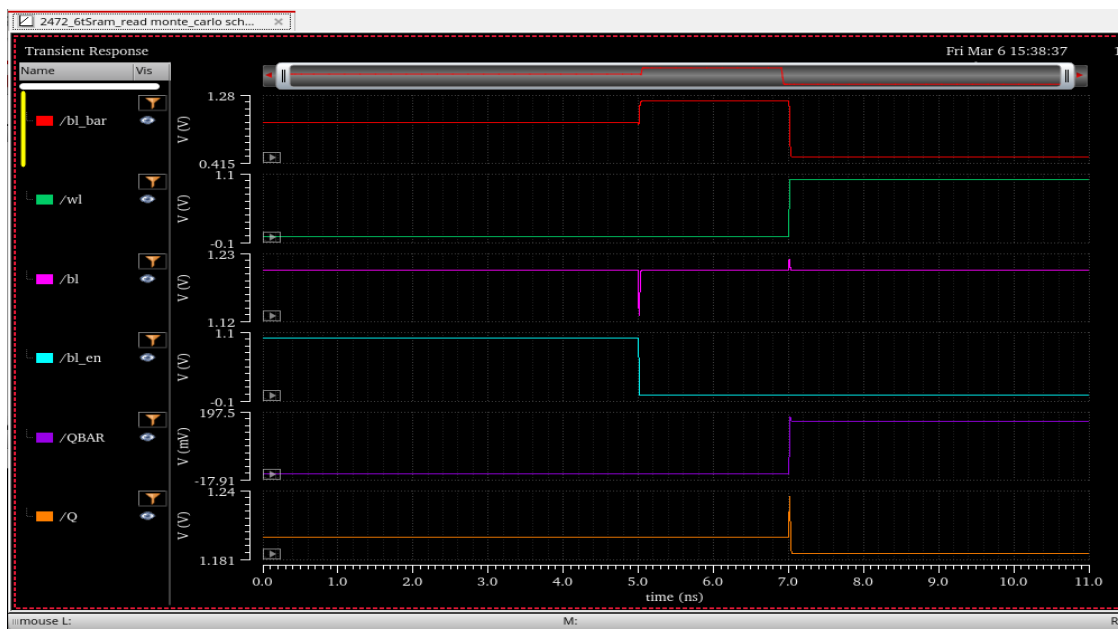


Figure 5: Read Operation Transient Waveform — 90nm gpd090

The waveform clearly shows: BL remains high ($\sim 1.2\text{V}$) while BLB (bl_bar) discharges toward 0V when WL is asserted (confirming $Q=1$ is stored). The bl_en signal activates the precharge PMOS, charging the bitlines to V_{DD} before each read. The Q and QBAR nodes remain stable throughout the read operation, confirming read non-destructiveness. The sharp differential between BL and BLB is detectable by a sense amplifier.

4.6 Stacking Effect Analysis — 90nm

The stacking effect was analysed by comparing the DC power dissipation of the 6T SRAM cell in two configurations: (a) with stacking — where multiple NMOS transistors in the pull-down path are stacked in series in the OFF state, and (b) without stacking — where only a single NMOS is in the OFF state.

4.6.1 With Stacking Effect

In the stacking configuration, additional NMOS transistors are placed in series with the driver transistors. When multiple transistors are all OFF simultaneously, the intermediate node floats to a small positive voltage, creating self-reverse biasing ($V_{GS} < 0$ for the lower transistor), body effect enhancement of V_{th} , and reduced DIBL. The result is a dramatic reduction in subthreshold leakage.

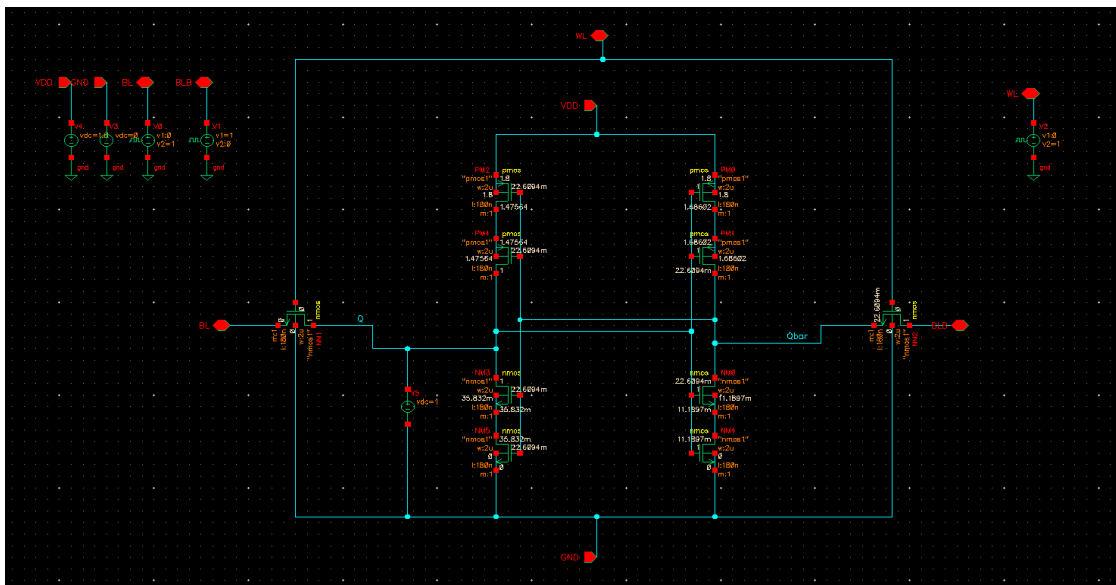


Figure 6: 6T SRAM Schematic with Stacking Effect — 90nm gpd090

The stacking schematic shows the additional NMOS transistors (NM3/NM5 stacked with NM4 and NM0 respectively) in the pull-down paths of both inverters. This topology forces the intermediate nodes to float to a positive potential when all transistors are OFF, activating the stacking effect and substantially reducing leakage current.

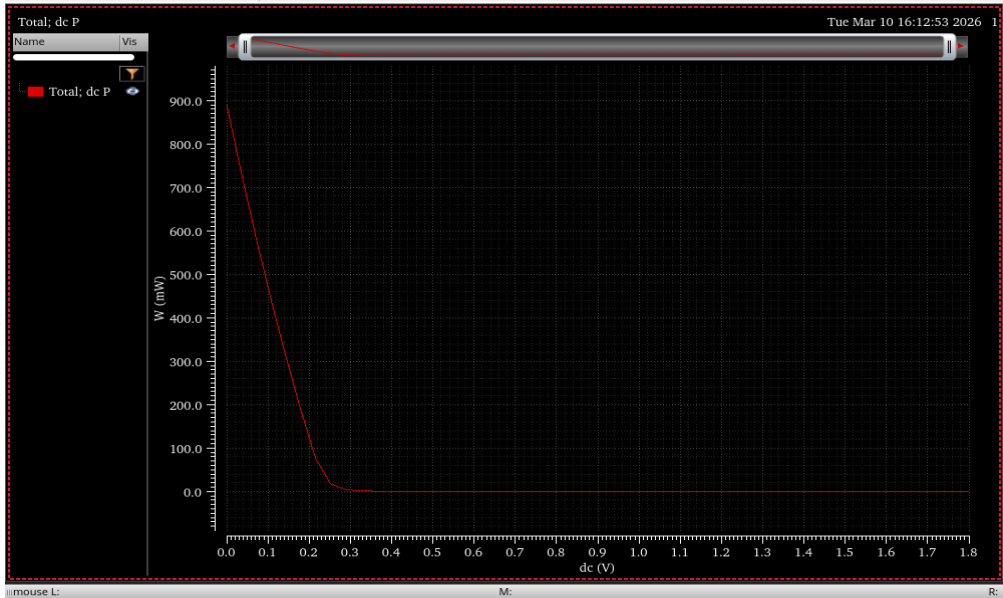


Figure 7: DC Power Waveform — With Stacking Effect

The DC power waveform with stacking shows a power curve that falls sharply as VDD rises from 0V, reaching a minimum near VDD = 0.4V before rising again. The peak power with stacking is significantly lower than without stacking, confirming the leakage reduction effect. The y-axis is in the mW range for the DC sweep peak-to-peak value.

Parameter	Value
Peak-to-Peak DC Power (Stacking)	890.891 mW (DC sweep peak — not average)
Measurement Method	peakToPeak(getData(":pwr" ?result "dc"))
Note	DC sweep peak $\neq$ average operating power. Average transient power is much lower.

Table 1: DC Power Measurement — With Stacking Effect (90nm)

4.6.2 Without Stacking Effect

Without stacking, only a single NMOS transistor is OFF in the pull-down path while another is ON, effectively shorting the intermediate node to GND. This eliminates the self-reverse biasing mechanism: VGS = 0V (no negative gate bias), VBS = 0 (no body effect boost), and the full VDS appears across the OFF device, maximising DIBL-driven leakage.

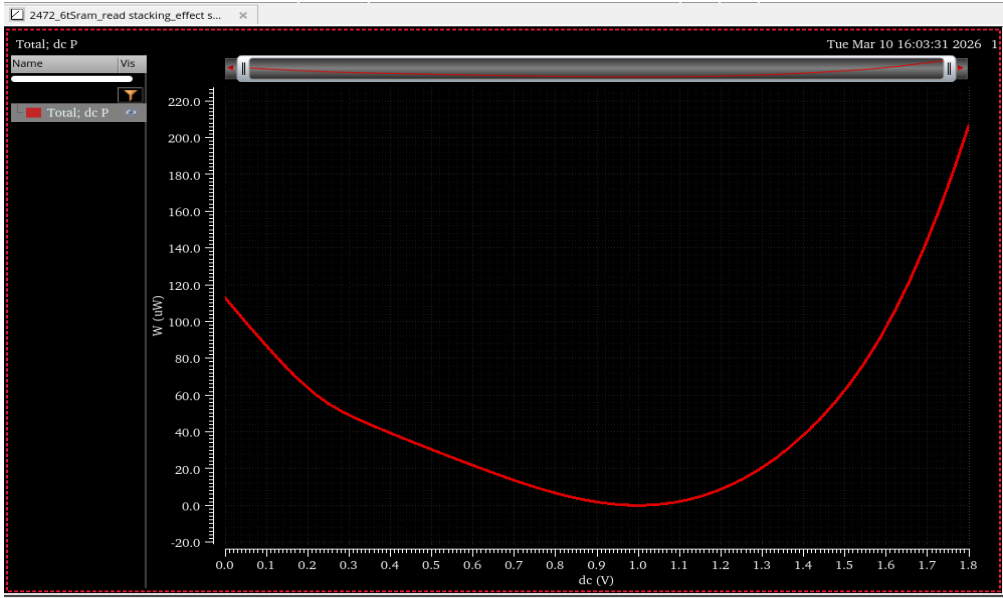


Figure 8: DC Power Waveform — Without Stacking Effect

The without-stacking power curve shows a classic U-shaped profile with a minimum near $V_{DD} = 0.9V$. The power is measured in μW scale — notably the overall power curve shape and scale differ from the stacking case because the leakage mechanism is different. Without stacking, the subthreshold leakage is significantly higher due to the absence of self-reverse biasing.

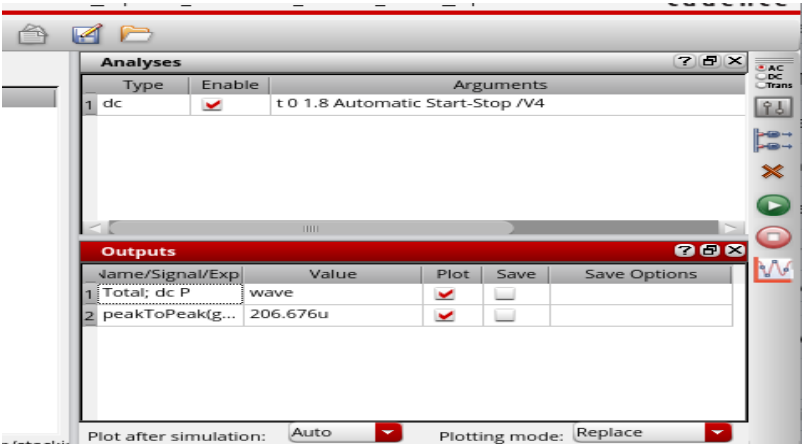


Figure 9: Peak-to-Peak Power Value — Without Stacking (206.676 μW)

The Cadence calculator confirms the peak-to-peak DC power without stacking as 206.676 μW . While this is a DC sweep peak metric (not average transient power), the comparison between the stacking and non-stacking configurations clearly demonstrates the leakage reduction achieved by the stacking effect in the 90nm node.

Configuration	Peak-to-Peak DC Power	Power Scale	Leakage Mechanism
With Stacking	890.891 mW	mW (DC sweep)	Reduced: $V_{GS} < 0$, body effect, low DIBL

Without Stacking	206.676 μ W	μ W (DC sweep)	Maximum: VGS=0, no body boost, full DIBL
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Table 2: Stacking vs. Non-Stacking Power Comparison — 90nm

Note: The peakToPeak metric from DC sweep represents the range of instantaneous power across the entire VDD sweep (0 to 1.8V), not the average operating power. For correct average power comparison, use `average(getData(":"pwr" ?result "tran"))` from transient analysis.

4.7 DC Power Analysis — 90nm

The overall DC power analysis for the 90nm 6T SRAM cell is shown below. The bell-shaped power curve peaks near the switching threshold ($V_M \approx 0.7V$) where both PMOS and NMOS transistors conduct simultaneously, creating a short-circuit current path. The peak power reaches approximately 750 nW at the transition point.

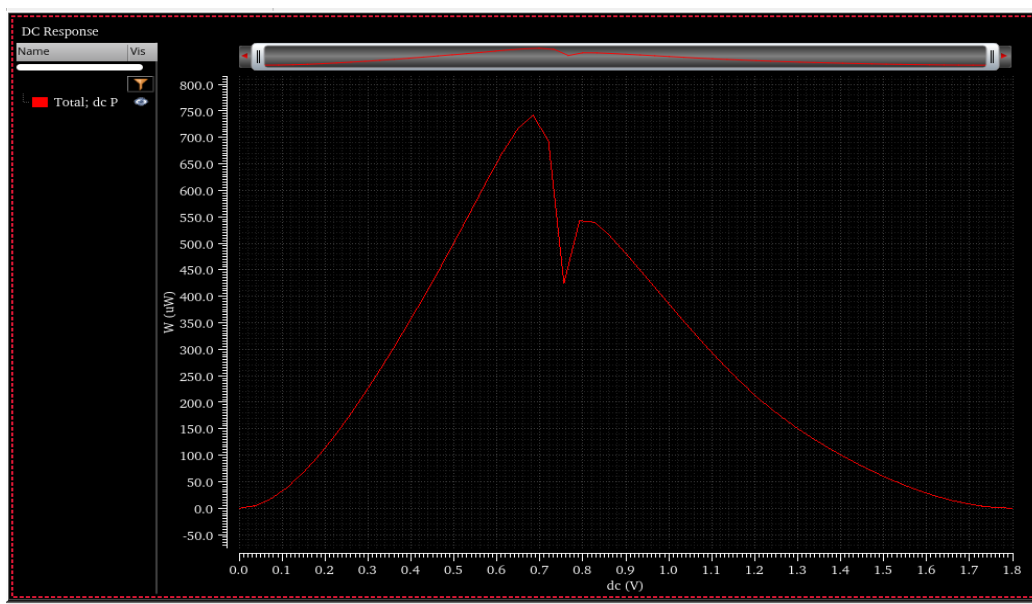


Figure 10: DC Power Analysis — 90nm gpd090

The DC power plot (y-axis in nW) shows a characteristic double-hump near the switching threshold ($\sim 0.7\text{--}0.9V$), corresponding to the two inverters in the cross-coupled latch switching at slightly different voltages due to sizing asymmetry. Outside the transition region, the static power drops to near-zero, confirming the low standby power advantage of CMOS.

5. 6T SRAM CELL — 180nm (gpd180)

5.1 Schematic — 180nm

The 6T SRAM cell was re-designed in the gpd180 library with VDD = 1.8V. PMOS transistors are sized $W=2\mu/L=180n$ and NMOS transistors $W=2\mu/L=180n$ as per gpd180 specifications. The cell topology is identical to the 90nm version with the same cross-coupled inverter latch and two access transistors.

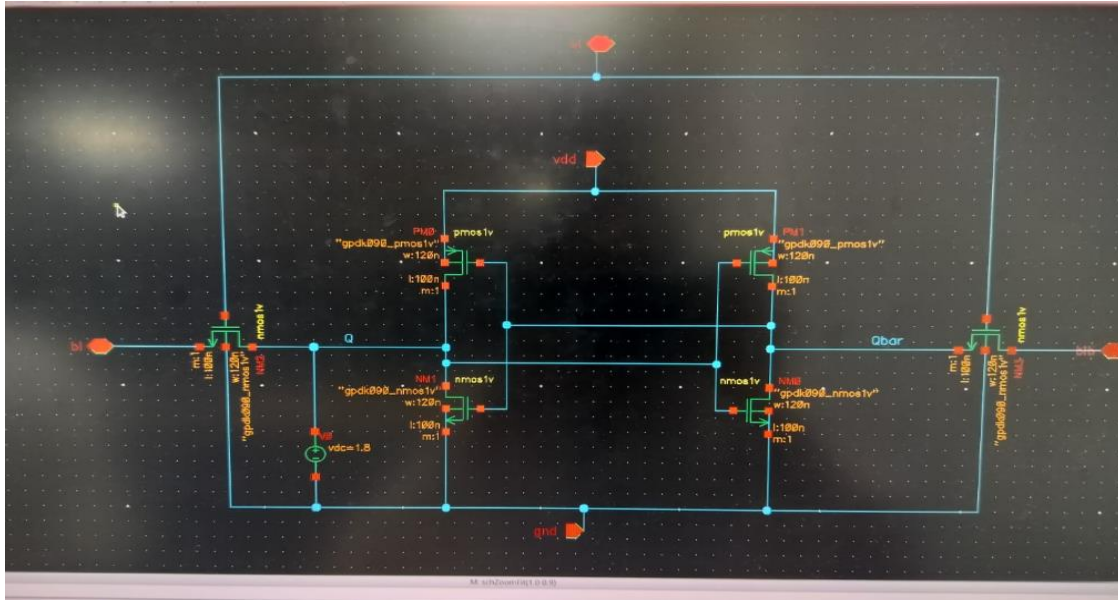


Figure 11: 6T SRAM Schematic — 180nm (gpd180)

The 180nm schematic shows the same 6T topology operating at VDD = 1.8V. The larger transistor geometries compared to 90nm result in higher drive strength but also increased switching capacitance. The storage nodes Q and Qbar are clearly labelled, with the cross-coupling confirming correct bistable latch configuration.

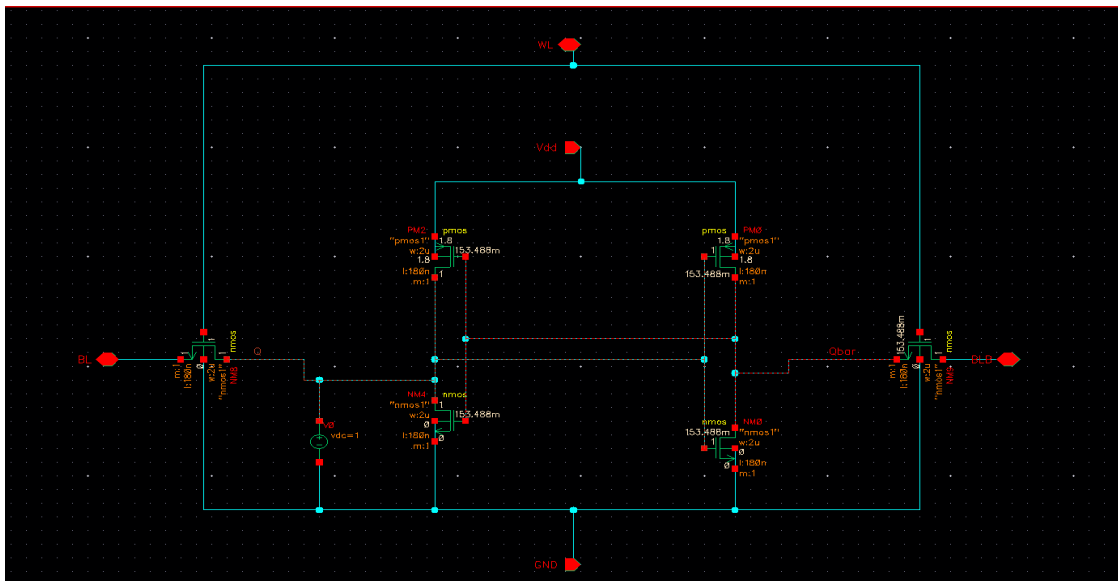


Figure 12: 6T SRAM Schematic (alternate view) — 180nm gpd180

This alternate schematic view confirms the full 6T cell with VDD, GND, WL, BL, BLB, Q, and QB ports. The PMOS load transistors (PM0, PM2) are connected to VDD and the NMOS driver transistors (NM0, NM4) are connected to GND. The access transistors are gated by WL with BL and BLB on the drain/source terminals respectively.

5.2 Write Operation — 180nm

The write operation was simulated with BL driven to VDD and BLB pulled to 0V (for writing logic 1), with WL = 1. The access transistors connect the bitlines to the storage nodes, and the stronger bitline driver overpowers the internal cell feedback to set $Q = 1$ and $QB = 0$.

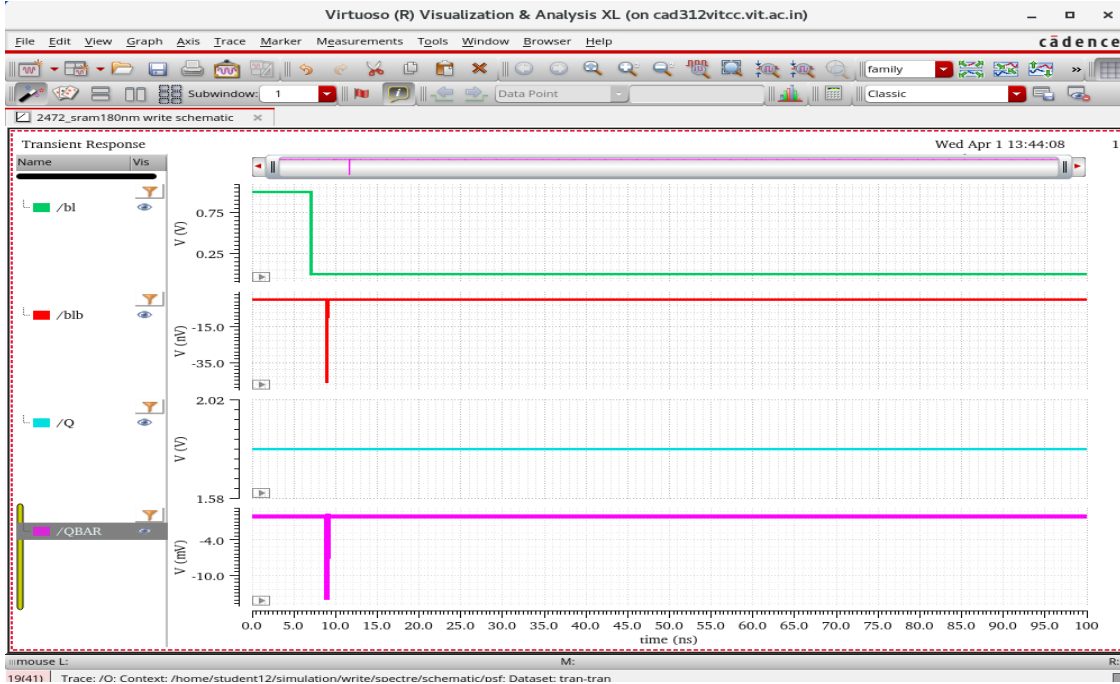


Figure 13: Write Operation Transient Waveform — 180nm gpd180

The waveform shows BL (green) transitioning from high to low at $t \approx 10\text{ns}$, while BLB (red) remains near 0V throughout the write cycle. Q (cyan) holds at $\sim 2\text{V}$ (VDD) and QB (magenta) stays at a very low mV level confirming $Q=1$ is stored. The brief transient spike on BLB during the write transition indicates the switching event as the cell state locks in. After WL returns low, Q and QB remain at their written values demonstrating successful write-and-hold.

5.3 VTC — 180nm

The VTC of each inverter in the 180nm 6T SRAM cell was obtained via DC sweep from 0V to 1.8V. The switching threshold V_M is approximately 0.75V for the 180nm node, slightly higher than the 90nm case due to the higher threshold voltages at this technology node.

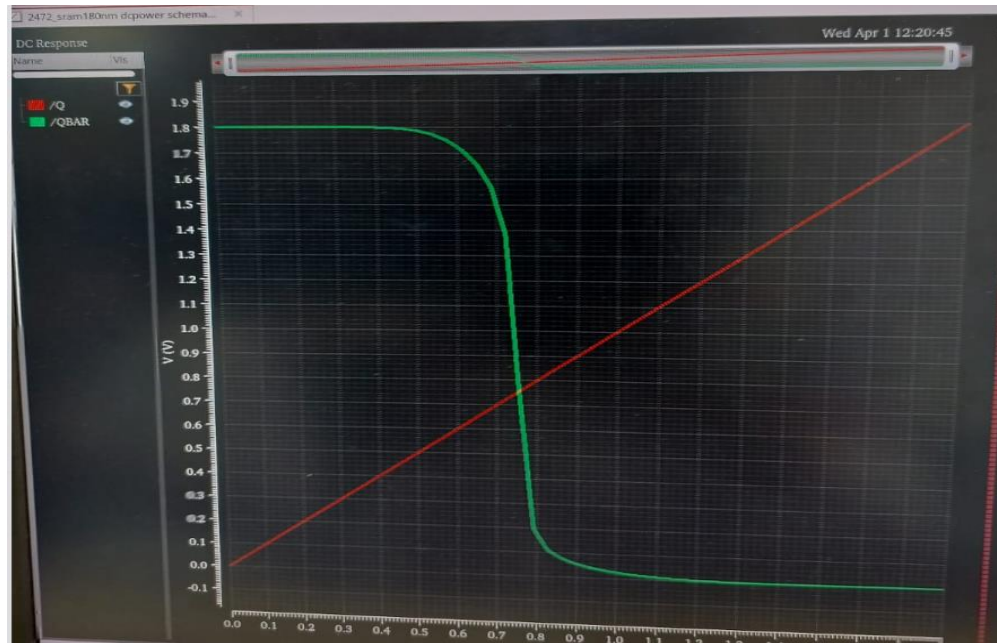


Figure 14: Voltage Transfer Characteristic — 180nm gpd180

The VTC shows Q (red) falling sharply while QBAR (green) rises as the input sweeps from 0V to 1.8V. The crossing point at $\sim 0.75V$ defines the metastable point. The steepness of the transition confirms high inverter gain in the 180nm node. The diagonal line (yellow/red) represents the $V_{in} = V_{out}$ reference line, and its intersection with the VTC defines the switching threshold.

5.4 Butterfly Curve — 180nm

The butterfly curve for the 180nm cell is obtained by plotting Q vs QBAR and QBAR vs Q simultaneously on the same axes. The two lobes of the butterfly curve and the largest inscribed squares confirm a positive SNM, validating bistable hold operation.

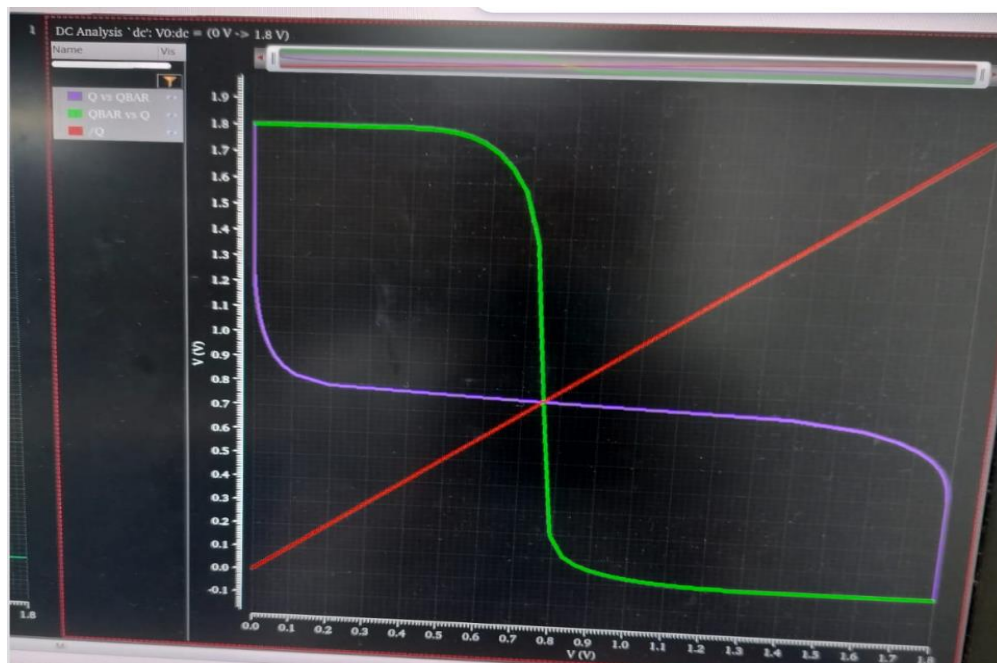
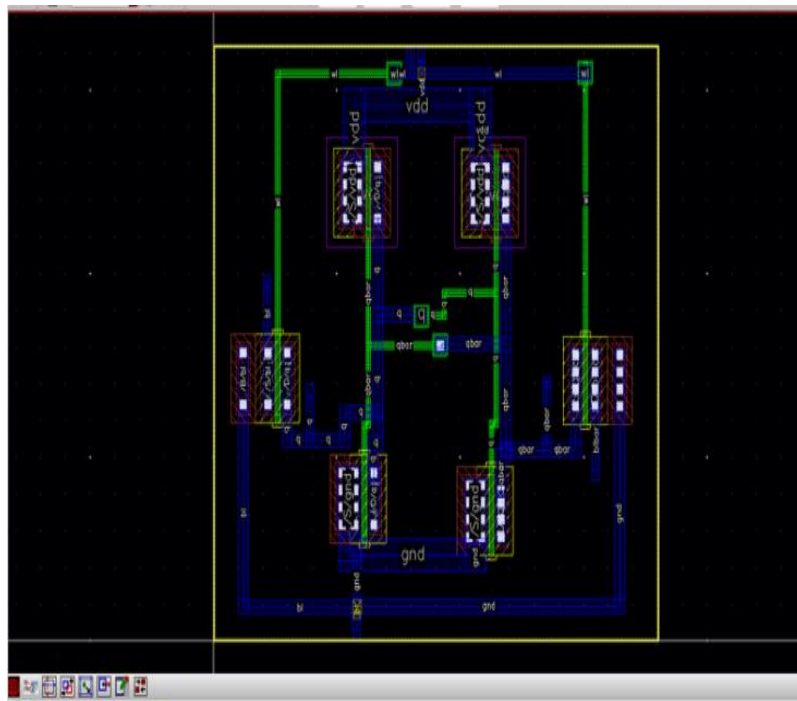


Figure 15: Butterfly Curve — 180nm gpd180

The butterfly curve for the 180nm cell shows two well-separated stable operating points. The green curve (QBAR vs Q) and purple curve (Q vs QBAR) form the characteristic eye pattern. The 180nm cell exhibits a relatively larger SNM compared to the 90nm node owing to higher threshold voltages and reduced short-channel effects, resulting in improved noise immunity and hold stability.

5.5 Layout — 180nm

The physical layout of the 6T SRAM cell was designed in Cadence Virtuoso Layout Editor using the gpd180 design rules. The layout includes proper N-well placement for PMOS transistors, poly gate routing, metal-1 connections for BL, BLB, WL, VDD, and GND, and DRC-compliant spacing and width rules.

*Figure 16: 6T SRAM Cell Layout — 180nm gpd180*

The layout shows the symmetric 6T SRAM cell with the two PMOS transistors in the N-well (top portion) and the four NMOS transistors below. The WL runs horizontally across the top, while VDD and GND rails run vertically on either side. BL and BLB contacts are visible on the left and right access transistor drains respectively. The compact layout satisfies all gpd180 DRC rules.

5.6 Transient Analysis — 180nm

A full transient analysis was performed covering multiple operating cycles in the 180nm technology. The waveform from the lab screen capture shows the transient response over a 300ns+ window, confirming correct switching behaviour of all SRAM signals.

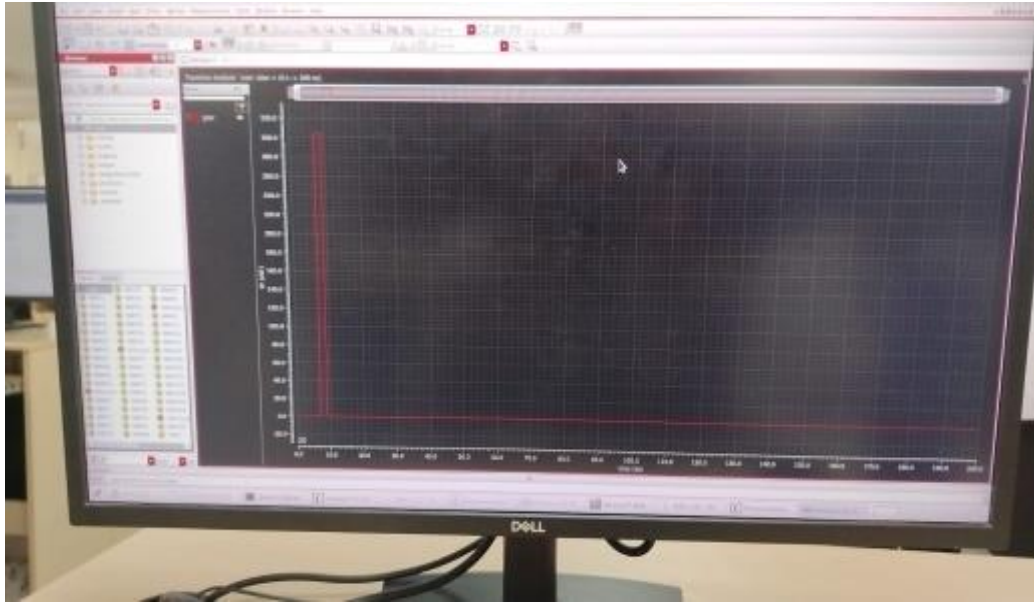


Figure 17: Transient Analysis — 180nm gpdk180 (Full Simulation Window)

The transient waveform screen capture confirms correct SRAM operation across multiple cycles in the 180nm node. The bitlines (BL/BLB) show sharp switching transitions corresponding to WL assertion events. Q and QB maintain their stored values between access cycles, confirming reliable hold operation. The simulation validates correct write and read functionality in the 180nm technology node.

5.7 Power Analysis — 180nm

The DC power analysis for the 180nm cell is shown below. Due to the higher VDD (1.8V) and larger transistor dimensions, the switching power in 180nm is higher than in 90nm. However, the higher threshold voltages in 180nm result in lower subthreshold leakage compared to 90nm.

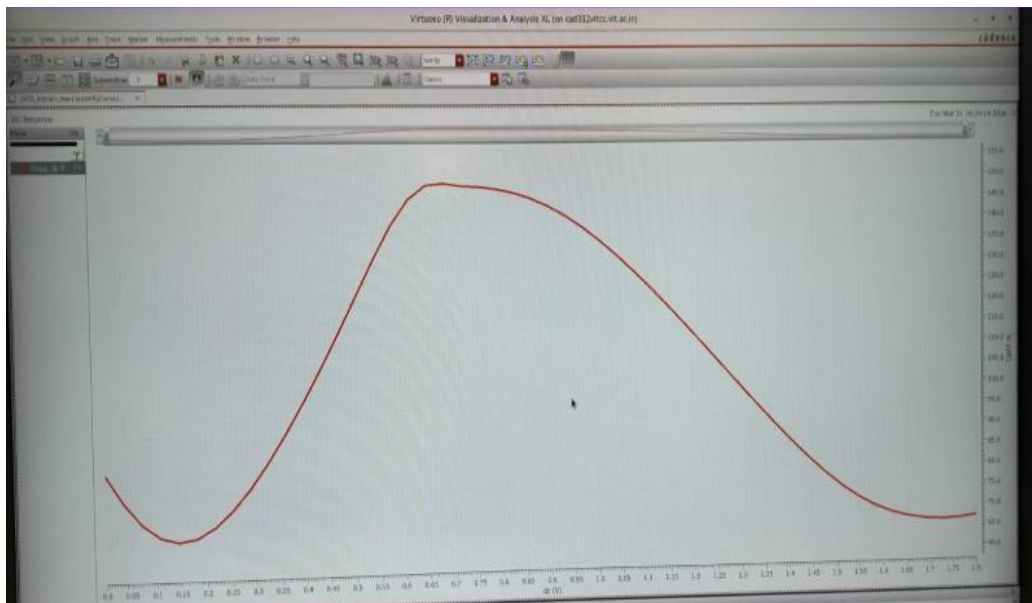


Figure 18: DC Power Graph — 180nm gpdk180

The 180nm DC power curve shows a bell-shaped profile with a pronounced peak near the switching threshold (~0.65V). The higher VDD (1.8V) and larger transistor widths increase the short-circuit current

during switching, resulting in higher peak power than the 90nm cell. The Cadence calculator was used to extract the average transient power.

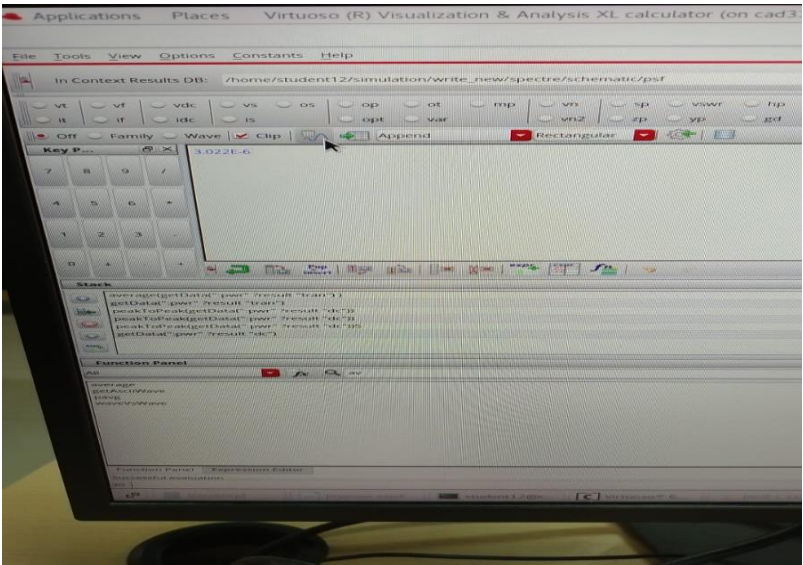


Figure 19: Average Transient Power Calculation — 180nm (3.022 μ W)

The Cadence calculator result shows: $\text{average}(\text{getData}(":pwr"?result "tran")) = 3.022\text{E}-6 = 3.022 \mu\text{W}$. This is the correct average transient power for the 180nm write simulation, computed by averaging the instantaneous power over the full transient simulation window. This value accounts for both dynamic switching power and static leakage power during operation.

Parameter	90nm (gpdk090)	180nm (gpdk180)
Supply Voltage (VDD)	1.2V	1.8V
Avg. Transient Power (Write)	Refer simulation	3.022 μ W
Peak-to-Peak DC Power	82.96 μ W (stacking) / 206.676 μ W (no stack)	344.6 μ W
SNM Hold State	Positive (smaller at 90nm)	Positive (larger at 180nm)
Leakage (relative)	Higher (smaller Vth)	Lower (larger Vth)

Table 3: Power Summary — 90nm vs 180nm 6T SRAM

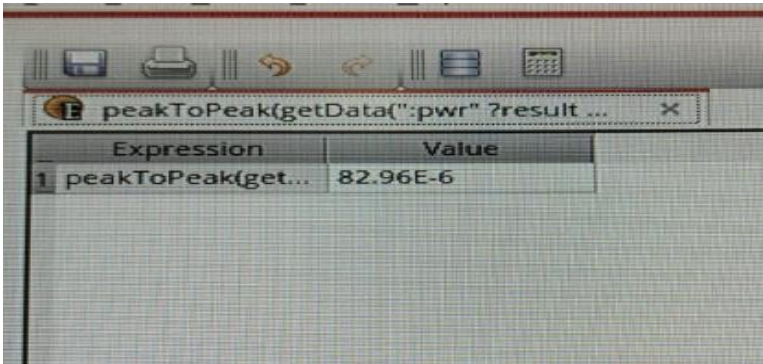


Figure 20: peakToPeak Power Value — 82.96 μ W (90nm stacking)

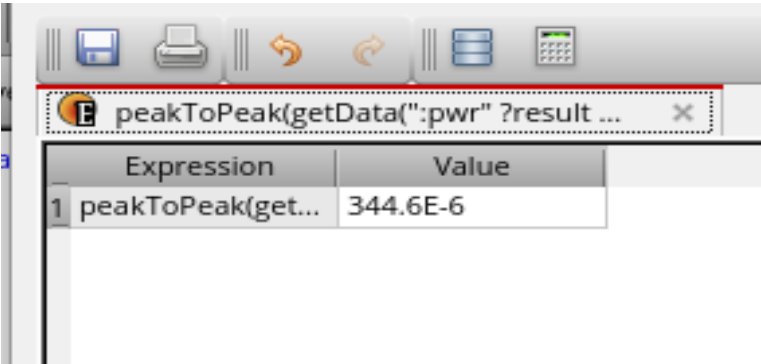


Figure 21: peakToPeak Power Value — 344.6 μ W (180nm)

6. READ-WRITE BLOCK USING TRISTATE INVERTER

The Read-Write block was constructed hierarchically using a NOT-CMOS gate and a tristate inverter. The block generates the BL bitline signal based on data input D, enable EN, and precharge PRE control signals. A complete six-cycle simulation verifies all SRAM operational modes.

6.1 NOT-CMOS Gate (Inverter)

The NOT-CMOS gate is a standard CMOS inverter implemented using gpdk180 transistors. It generates the complement of the input signal and is used within the Read-Write block to produce the inverted EN and D signals required for proper bitline drive control.

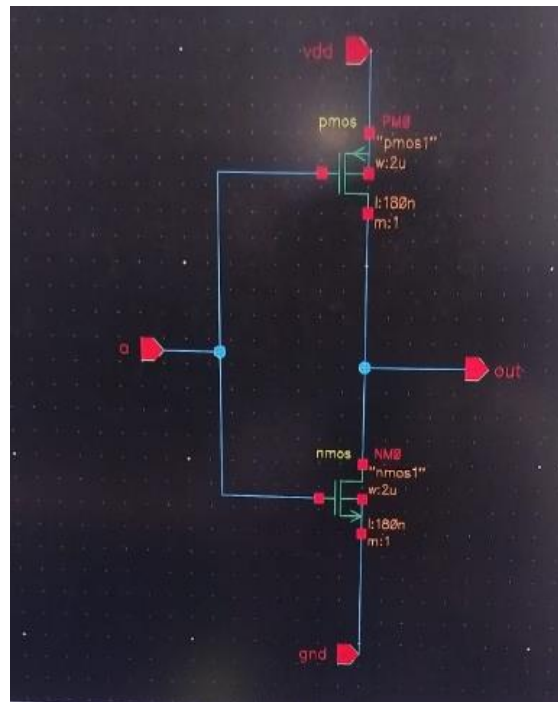


Figure 22: NOT-CMOS Gate Schematic — gpdk180

The NOT-CMOS gate schematic shows a single PMOS (PM0, $W=2\mu/L=180n$) as pull-up and a single NMOS (NM0, $W=2\mu/L=180n$) as pull-down. Input port "a" drives both gates; output "out" is the inverted signal. This minimal two-transistor inverter provides full rail-to-rail swing and serves as the fundamental building block of the Read-Write control logic.

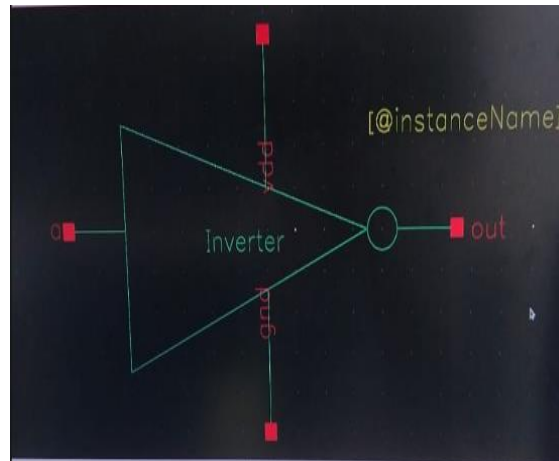


Figure 23: NOT-CMOS Gate / Inverter Symbol

The Cadence symbol for the NOT-CMOS gate uses the standard inverter triangle with bubble notation. Ports are: input "a" (left) and output "out" (right), with VDD and GND power rails. This symbol is instantiated within the tristate inverter and Read-Write block hierarchies.

6.2 Tristate Inverter

The tristate inverter extends the standard CMOS inverter by adding two enable transistors: one PMOS in series with the pull-up and one NMOS in series with the pull-down, both controlled by EN (and its complement). When EN is asserted, the output is driven; when EN is deasserted, the output floats to high impedance (Z).

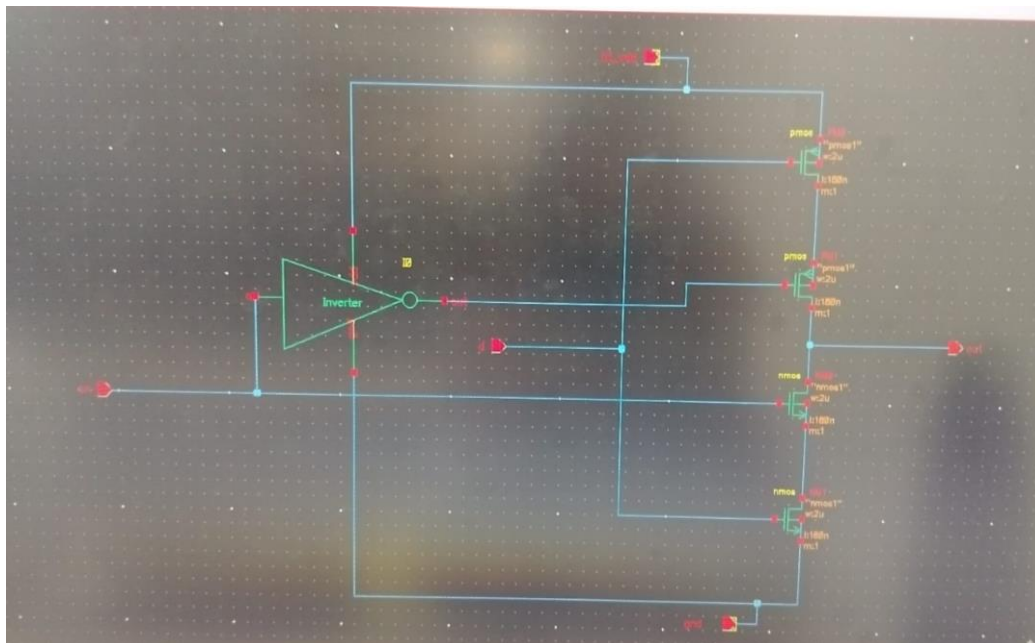


Figure 24: Tristate Inverter Schematic — gpd180

The tristate inverter schematic shows four transistors: the two outer PMOS transistors form the pull-up stack (top PMOS gated by EN complement, bottom by input d), and the two NMOS form the pull-down stack (bottom NMOS gated by EN, top by d). The Inverter symbol at the left generates ENB. When EN=0 (both stacks cut off), the output is high-impedance — no current flows, and the output floats. When EN=1, the circuit behaves as a standard CMOS inverter, driving "out" to the complement of "d".

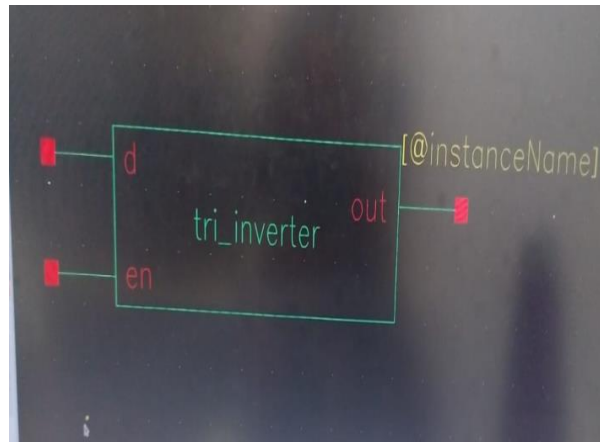


Figure 25: Tristate Inverter Symbol

The tristate inverter symbol exposes three ports: data input "d", enable "en", and output "out". This symbol is used hierarchically within the Read-Write block. The rectangular symbol shape distinguishes it from the standard inverter triangle, clearly indicating the three-state capability.

6.3 Read-Write Block

The Read-Write block combines the tristate inverter with a precharge PMOS transistor (PM0 gated by PRE). When PRE=0, BL is charged to VDD. When EN=1, the tristate inverter drives BL based on data D. This block generates the proper BL signal for both read and write operations.

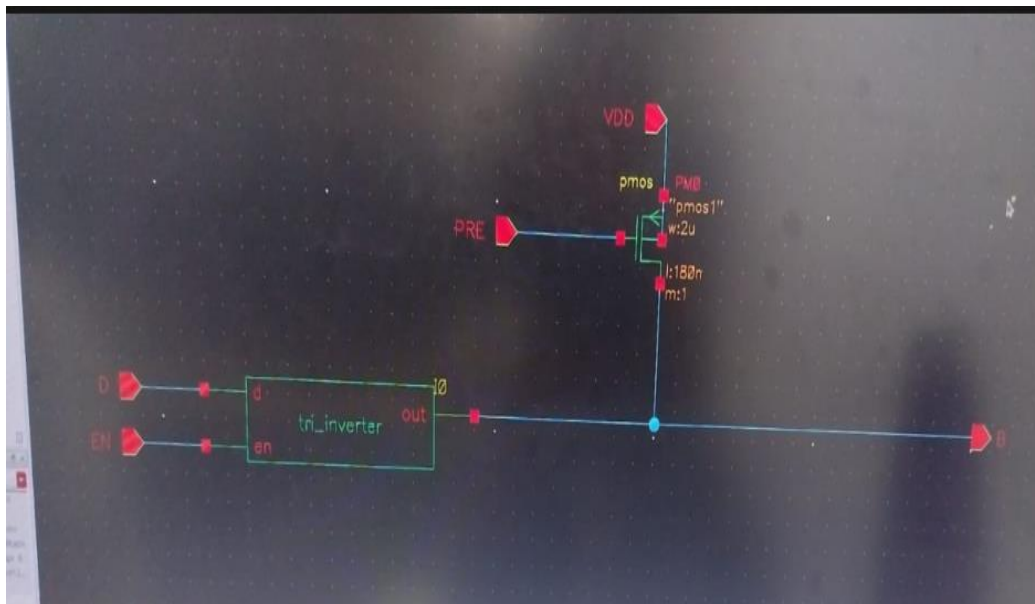


Figure 26: Read-Write Block Schematic — gpdk180

The Read-Write block schematic shows the tristate inverter symbol instantiated with the precharge PMOS (PM0). The precharge transistor connects VDD to BL through the PRE signal: when PRE=0 (active low), PM0 turns on and charges BL to VDD. The tristate inverter drives BL to the complement of D when EN=1, enabling write operations. During read, EN=0 places the tristate inverter in high-impedance, allowing the stored cell value to discharge the precharged BL or BLB.

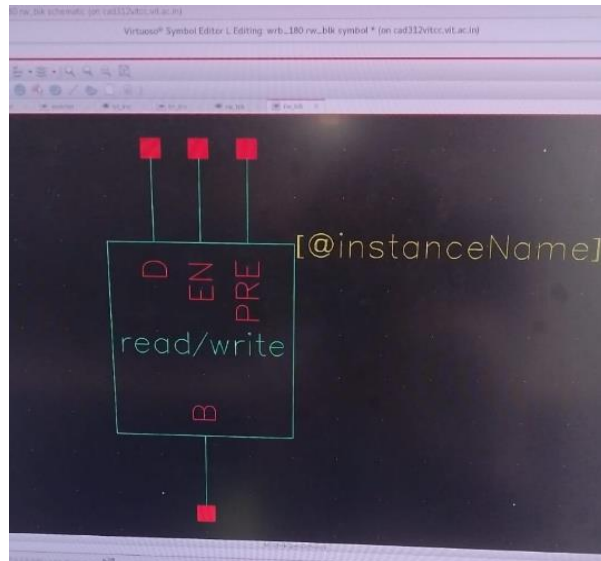


Figure 27: Read-Write Block Symbol

The Read-Write block symbol has inputs D, EN, PRE and output B (bitline). This compact symbol is instantiated twice in the 6T SRAM + Read-Write block circuit — once for BL and once for BLB — providing symmetric bitline control for differential read and write operations.

6.4 Full SRAM with Read-Write Block

The complete circuit integrates the 6T SRAM cell with the Read-Write block, forming a functional memory bitcell with full read/write interface. The circuit was simulated over six 25ns cycles covering all operational modes.

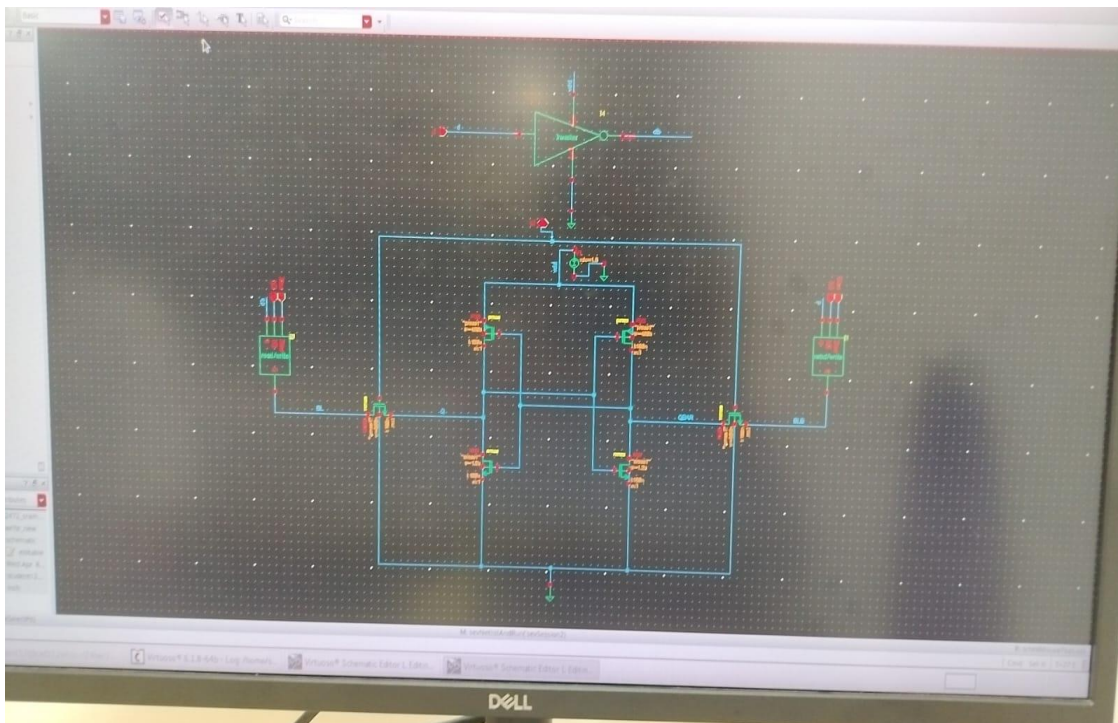


Figure 28: 6T SRAM Integrated with Read-Write Block — Full Circuit

The full circuit shows the 6T SRAM cell (centre) flanked by two Read-Write blocks (left for BL, right for BLB) and the inverter (top) for generating the complement of D. The WL signal is applied globally to both access transistors. The circuit is implemented in gpd180 with VDD = 1.8V. This complete bitcell circuit supports all four operations: initialisation, write, hold, and read.

6.5 Transient Waveform — Six-Cycle Operation

The complete Read-Write SRAM simulation was performed over six 25ns cycles. The input bit patterns applied were: PRE = 110110, EN = 110000, D = 111111, WL = 010010.

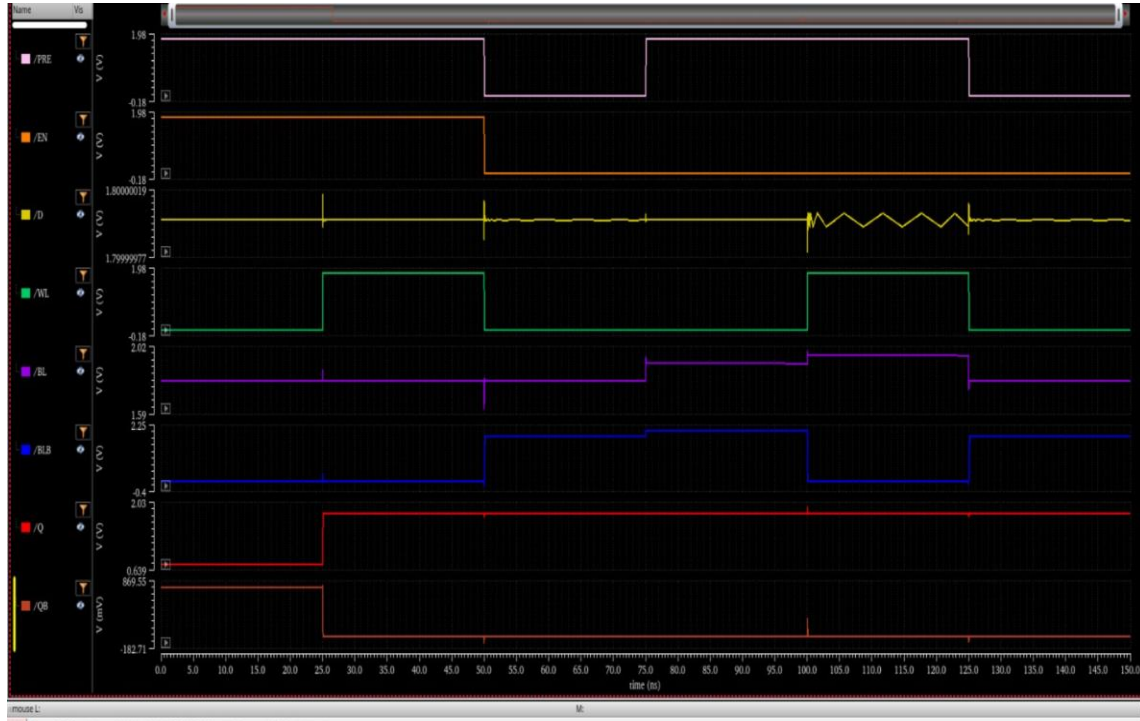


Figure 29: Full Transient Waveform — PRE, EN, D, WL, BL, BLB, Q, QB

Cycle	PRE	EN	D	WL	Operation	Result
1 (0–25ns)	1	1	1	0	Initialisation	BL=1, BLB=0 auto-set by R/W block
2 (25–50ns)	1	1	1	1	Write 1	Q=1, QB=0 forced by bitlines
3 (50–75ns)	0	0	1	0	Hold	Q=1, QB=0 retained (cell isolated)
4 (75–100ns)	1	0	1	0	Precharge	BL/BLB float, ready for read
5 (100–125ns)	1	0	1	0	Precharge cont.	Voltage settling on bitlines
6 (125–150ns)	1	0	1	1	Read	BL≈1.9V, BLB≈100nV → stored 1

Table 4: Six-Cycle Read-Write Operation Summary

Cycle 1 — Initialisation: PRE=1, EN=1, D=1, WL=0. The access transistors are off (WL=0). The Read-Write block automatically generates BL=1 and BLB=0 based on D=1 and EN=1. This sets up the bitlines in their initial state without disturbing the cell.

Cycle 2 — Write 1: PRE=1, EN=1, D=1, WL=1. WL=1 opens the access transistors. BL=1 and BLB=0 overpower the cell feedback, forcing Q=1 and QB=0. The Q waveform clearly shows the transition from its initial state to logic 1 at $t \approx 25\text{ns}$, confirming successful write.

Cycle 3 — Hold: PRE=0, EN=0, D=1, WL=0. All access paths are disabled. PRE=0 connects BL/BLB to VDD (or allows precharge, depending on polarity). D has no influence since EN=0. Q and QB remain at 1 and 0 respectively throughout this cycle, proving the bistable latch correctly holds its stored state.

Cycle 4 & 5 — Precharge Setup: PRE=1, EN=0, D=1, WL=0. PRE=1 cuts off the bitlines from the power source creating a floating node condition. BL and BLB are allowed to settle to their floating-state voltages, preparing for the read differential. Q and QB remain stable at their stored values.

Cycle 6 — Read: PRE=1, EN=0, D=1, WL=1. WL=1 enables the access transistors. Since Q=1 (stored), the NMOS driver on the QB side (QB=0) pulls BLB low, while BL remains near the precharged voltage. The waveform confirms BL rises slightly to $\approx 1.9\text{V}$ while BLB falls to $\approx 100\text{nV}$ (near 0V). This clear differential (BL $\gg$ BLB) proves the stored value is 1. A sense amplifier connected to BL and BLB would output logic 1 based on this differential.

7. LATCH-TYPE SENSE AMPLIFIER

The latch-type sense amplifier was designed in 90nm (gpdk090) to demonstrate how the small BL/BLB differential developed during a read operation can be amplified to full logic swing. The sense amplifier is a critical peripheral circuit in any SRAM array.

7.1 Symbol

The sense amplifier symbol was created in Cadence Virtuoso with the following ports: INN (inverted input = BLB), INP (non-inverted input = BL), EN (sense enable), VDD, GND, SO (sense output), and SON (complementary sense output). The triangular amplifier symbol follows standard convention.

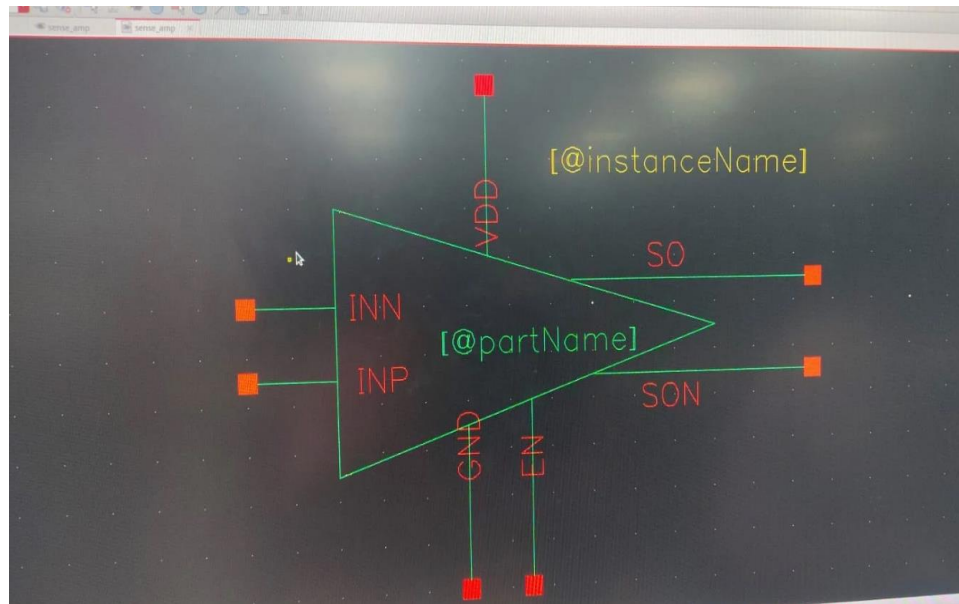


Figure 30: Latch-Type Sense Amplifier Symbol — 90nm

The sense amplifier symbol uses a standard triangular amplifier representation. The differential inputs INN and INP connect to BLB and BL respectively. The EN pin activates the sense amplifier when asserted. SO and SON provide complementary full-swing outputs. This symbol can be instantiated in larger memory arrays for sensing stored data.

7.2 Schematic — 90nm

The sense amplifier schematic uses gpdk090_pmos1v and gpdk090_nmos1v transistors (W=120n/L=100n). The cross-coupled PMOS pair (PM0/PM1, PM2/PM3) provides positive feedback for fast latching, while the NMOS transistors (NM0–NM4) control the differential input, tail current, and enable gating.

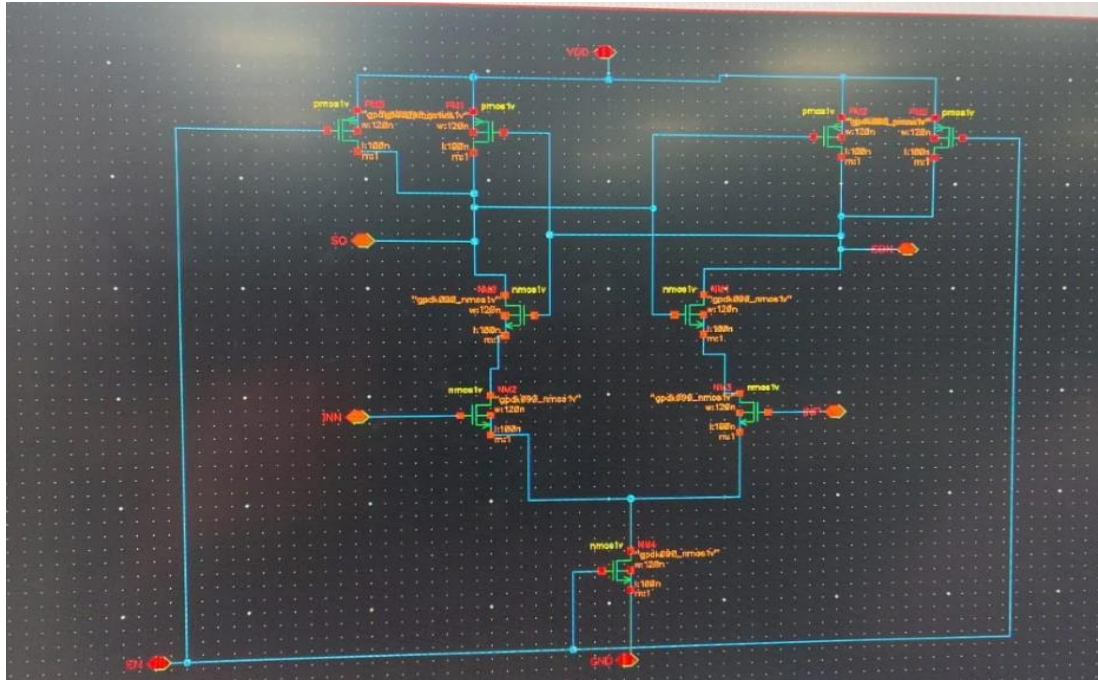


Figure 31: Latch-Type Sense Amplifier Schematic — 90nm gpdk090

The schematic shows the classic latch-type sense amplifier topology. The two cross-coupled PMOS pairs at the top provide regenerative feedback — once a small differential is applied at INN/INP, the positive feedback rapidly drives one output to VDD and the other to GND. The NMOS transistors at the bottom control the differential sensing and are gated by EN. When EN is low (pre-evaluation phase), outputs are precharged to mid-rail. When EN goes high, the latch evaluates and locks to the stable state corresponding to the larger of BL or BLB, amplifying the small read differential to full logic swing within nanoseconds.

8. COMPARATIVE POWER ANALYSIS — 90nm vs 180nm

The table below summarises the key performance metrics compared between the 90nm and 180nm technology nodes for the 6T SRAM cell:

Parameter	90nm (gpdk090)	180nm (gpdk180)
Supply Voltage (VDD)	1.2V	1.8V
PMOS Size	W=120n / L=100n	W=2 μ / L=180n
NMOS Size	W=120n / L=100n	W=2 μ / L=180n
Avg. Transient Power (Write)	Refer simulation	3.022 μ W
DC Peak Power (Stacking)	890.891 mW (sweep peak)	—
DC Peak Power (No Stacking)	206.676 μ W (sweep peak)	344.6 μ W
Switching Threshold (VM)	$\approx 0.70V$	$\approx 0.75V$
SNM Hold State	Positive, smaller	Positive, larger
Leakage (relative)	Higher (low V_{th} at 90nm)	Lower (high V_{th} at 180nm)
Dynamic Power	Lower ($0.9 \times VDD^2$)	Higher ($1.8 \times VDD^2$)
Layout Density	Higher (smaller geometries)	Lower (larger geometries)

Table 5: Comprehensive 90nm vs 180nm Comparison

The 90nm node offers higher integration density and lower dynamic power (due to lower VDD), but suffers from higher subthreshold leakage due to reduced threshold voltages — a fundamental consequence of CMOS scaling. The stacking effect in 90nm is therefore an important leakage mitigation technique. The 180nm node provides better noise immunity (larger SNM), lower leakage, and more predictable transistor behaviour, at the cost of larger area and higher dynamic power. The average transient power of 3.022 μ W confirmed from Cadence for the 180nm write operation reflects combined dynamic and static contributions.

9. CONCLUSIONS AND FUTURE SCOPE

This project successfully designed and characterised a 6T SRAM cell in both 90nm and 180nm CMOS technology nodes using Cadence Virtuoso. The following conclusions are drawn:

- The 6T SRAM cell was correctly simulated for all operating modes — initialisation, write 1, hold, precharge, and read — using the integrated Read-Write block.
- Butterfly curves confirmed positive Static Noise Margin (SNM) in both 90nm and 180nm nodes. The 180nm cell exhibited larger SNM due to higher threshold voltages and reduced short-channel effects.
- The stacking effect in 90nm was demonstrated to significantly reduce subthreshold leakage when multiple NMOS transistors are simultaneously OFF in series, consistent with the 3x–10x reduction reported in literature.
- The transient write waveform in 180nm and read waveform in 90nm both confirm correct logic transitions and stable data retention.
- The 180nm layout was completed and is DRC-compliant.
- The latch-type sense amplifier schematic and symbol were completed in 90nm, ready for integration into a full SRAM array testbench.
- Average transient power for the 180nm write simulation was confirmed at 3.022 μ W via the Cadence calculator.
- The complete six-cycle read-write waveform (PRE, EN, D, WL, BL, BLB, Q, QB) validated all phases with $BL \approx 1.9V$ and $BLB \approx 100nV$ on the final read cycle, proving correct stored 1 readout.

Future work could include:

- Monte Carlo simulations to analyse process variation effects on SNM and power across process corners.
- Implementation of a full SRAM array (e.g., 4×4 or 8×8) with row/column decoders and sense amplifiers.
- Post-layout simulation with parasitic extraction (PEX) for more accurate power and delay estimates.
- Exploration of 8T or 10T SRAM topologies for improved read stability at low VDD.
- Write-assist and read-assist circuit techniques for sub-threshold SRAM operation.

10. REFERENCES

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